

Cross-lingual speaker verification for Sinhala and Tamil: a single-corpus benchmark with ablation isolation of transfer, calibration, and language-aware regularisation

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Abstract

We report the first published speaker-verification benchmark for Sri Lankan Sinhala and Tamil, languages absent from all public SV corpora and under-evaluated by even multilingual self-supervised encoders. Evaluation is two-tier: benchmark v0, a 527-speaker read-speech collection drawn from OpenSLR SLR52 (478 Sinhala speakers) and SLR65 (49 Tamil speakers) with 12,000 seed-reproducible trials per language, and benchmark v1, built on SLCeleb, our 280-speaker in-the-wild Sinhala/Tamil corpus (pending data placement). On this benchmark we compare training configurations — zero-shot transfer of VoxCeleb-trained checkpoints, training-free backend adaptation (adaptive symmetric score normalisation, PLDA, language-aware calibration), and in-language fine-tuning — and isolate each lever in a single-feature ablation. Per-language (monolingual Sinhala, monolingual Tamil) and cross-lingual (enrolment-test language mismatch) EERs are reported alongside pooled metrics. We describe the corpus-preparation pipeline, the deterministic reproducibility protocol, and the open-source toolkit released to support reproducible Sri Lankan SV research. The results characterise the cross-lingual EER gap for a single trained model and quantify each lever’s contribution to closing it.

Keywords: speaker verification; cross-lingual transfer; low-resource speech; Sinhala; Tamil; multi-task learning; score normalisation; PLDA; ECAPA-TDNN.

1 Introduction

Speaker verification (SV) — the task of deciding whether two recordings come from the same speaker — has achieved remarkable accuracy on English benchmarks: state-of-the-art systems report 0.8–1.5% EER on VoxCeleb1-O [4, 8]. The accuracy on lower-resource languages is less well characterised, particularly for Sinhala and Tamil, the two primary languages of Sri Lanka, which together account for approximately [N=22] million speakers but appear in no publicly available SV corpus.

The Sri Lankan deployment context exposes three additional sources of difficulty beyond pure label scarcity: *(i)* bilingual and code-switched speech is normal in everyday usage, so the deployment trial distribution includes pairs where enrolment and test languages differ; *(ii)* the dominant deployment channel is telephony, with attendant codec and bandwidth-restriction artefacts; and *(iii)* labelled-speaker collection is expensive at the scale a single research effort can muster, capping the corpus at the low hundreds of speakers.

This paper makes three contributions:

1. A reproducible two-tier Sinhala/Tamil SV benchmark — v0: 527 speakers (478 Sinhala from OpenSLR SLR52, 49 Tamil from OpenSLR SLR65 read speech) with 12,000 trials per

language; v1: SLCeleb, our 280-speaker in-the-wild corpus (pending) — with monolingual, cross-lingual, and pooled trial lists, generated deterministically from a single random seed.

2. Single-feature ablations isolating the contribution of cross-lingual fine-tuning, layer-wise LR decay, multi-task language-ID head, adaptive symmetric score normalisation, and PLDA scoring, with three-seed reproducibility.
3. An open-source toolkit [9] extending the upstream `voxceleb_trainer` codebase with eleven new features and twenty-six bug fixes, each selectable via a single configuration key.

The remainder of this paper is organised as follows. Section 2 situates the work in the broader SV and cross-lingual speech-processing literature. Section 3 describes the corpus, training, and evaluation protocol. Section 4 reports results. Section 5 discusses limitations and future work.

2 Related work

2.1 Speaker verification architectures and losses

ECAPA-TDNN [4] is the de-facto reference architecture for non-self-supervised SV, combining Res2Net-style multi-scale 1D convolutions, squeeze-and-excite channel re-weighting, multi-layer feature aggregation, and channel-attentive statistical pooling. AAM-Softmax [3] is the dominant embedding loss, producing discriminative representations on the hypersphere through a margin-augmented cosine softmax.

Self-supervised front-ends — WavLM [2], wav2vec 2 [1], mHuBERT-147 [10] — provide multi-lingual speech-aware representations that benefit low-resource SV but at significant parameter cost (95M–300M). Our toolkit ships infrastructure for both ECAPA-TDNN (paper’s primary encoder) and SSL front-ends (future work).

2.2 Cross-lingual and low-resource SV

When the target-language SV corpus is small ($\leq 10k$ utterances), training from scratch overfits. The standard remedy — cross-lingual fine-tuning from a high-resource (typically English VoxCeleb1) checkpoint with selective freezing and layer-wise learning-rate decay — is well established for ASR and translation but under-documented for SV-specific fine-tuning, motivating our explicit ablation.

2.3 Score normalisation and generative backends

Adaptive symmetric score normalisation (AS-Norm) [7] and two-covariance PLDA [11, 6] are eval-time backends widely used in NIST SRE pipelines. AS-Norm normalises trial scores against per-trial-side cohort statistics; PLDA replaces cosine scoring with a likelihood ratio under same- / different-speaker hypotheses. The two compose: PLDA scores can themselves be AS-Norm’d.

2.4 Multi-task and adversarial language-aware training

A multi-task language-ID head trained jointly with the speaker loss regularises the embedding to retain language-discriminative structure. Adversarial counterparts (DANN [5]) pursue the opposite goal: language-invariant embeddings. Both appear in the literature; we evaluate the multi-task variant as the primary lever and document DANN as infrastructure-ready future work.

3 Methodology

3.1 Corpus and trial construction

Evaluation is two-tier. Benchmark v0 comprises 527 speakers: 478 Sinhala from OpenSLR SLR52 (crowdsourced read speech, 16 kHz) and 49 Tamil from OpenSLR SLR65 (studio read speech, resampled to 16 kHz at load), keeping speakers with at least 10 utterances. Benchmark v1 is built on SLCeleb, our 280-speaker in-the-wild Sinhala/Tamil corpus (multi-genre, multi-session, with bilingual speakers), and is pending data placement. Within each speaker we hold out 20% of utterances for trial-pair construction; the remainder constitutes the training pool (31,049 training utterances at a 60-utterance-per-speaker cap for the fine-tuning experiments).

v0 trial pairs are sampled at a 1:3 target-to-impostor ratio at three operating points:

- Monolingual Sinhala: 3,000 target + 9,000 impostor pairs;
- Monolingual Tamil: 3,000 + 9,000 pairs;
- Cross-lingual (enrolment in one language, test in the other, for speakers present in both languages): empty at v0 — the OpenSLR corpora contain no bilingual speakers — and populated at v1 by SLCeleb’s bilingual speakers.

Pair sampling is reproducible from a single random seed (`tools/sl_dataprep.py -seed 42`).

3.2 Encoder, loss, and training

The speaker encoder is ECAPA-TDNN [4] with $C = 1024$ channels, 80 log-mel filterbanks (with pre-emphasis 0.97), embedding dimension 192, and channel-attentive statistical pooling. The training loss is AAM-Softmax [3] ($m = 0.2$, $s = 30$). Training augmentation reuses the MUSAN + RIR simulation chain with a fixed augmentation-probability distribution (0.30 clean / 0.20 reverb / 0.15 music / 0.15 speech / 0.20 noise).

3.3 Cross-lingual fine-tune protocol

Configurations P1 and P2 (Section 4) initialise from an ECAPA-TDNN checkpoint pretrained on VoxCeleb1 [8]. During SL fine-tuning the mel front-end (`torchfb`, `instancenorm`) is frozen, the global learning rate is scaled by $0.1\times$ from the from-scratch value, and layer-wise LR decay with $\gamma = 0.9$ is applied across the four ECAPA layers so that the layer closest to the speaker loss head receives the full LR and the input-side layer receives $0.9^3 \approx 0.73\times$ of it.

3.4 Multi-task language-ID training

A linear classifier $W_{\text{aux}} \in \mathbb{R}^{4 \times 192}$ maps the speaker embedding (pre-reshape, one prediction per utterance) to four language classes (Sinhala, Tamil, English, mix). The combined training loss is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{speaker}} + \lambda \mathcal{L}_{\text{lang}}, \quad \lambda = 0.3. \quad (1)$$

Speakers absent from the lookup carry `ignore_index = -1` and contribute no gradient. The lookup is generated automatically from the corpus directory layout by `tools/sl_dataprep.py`.

3.5 Eval-time scoring backends

Three eval-time scoring backends are compared:

1. Cosine similarity on L2-normalised embeddings.
2. Two-covariance PLDA [11] with LDA pre-reduction to $d = 200$ and length normalisation; closed-form fit on a held-out PLDA-train set extracted from the SL training data.

Table 1: Preliminary monolingual EER (%) on benchmark v0 (527 read-speech speakers; 12,000 trials per language). Fine-tuned rows are mean $\pm$ std over three seeds. Preliminary, benchmark v0: read speech, near single-session, closed-set speakers for the fine-tuned rows; v1 (SLCeleb) provides the open-set, in-the-wild condition.

System	Si	Ta
Zero-shot SpeechBrain ECAPA-TDNN	4.78	3.21
Zero-shot ReDimNet-B6	2.71	1.48
ReDimNet-B6 + AS-Norm (training-free)	2.07	0.90
WavLM-base+ LoRA fine-tune (3 seeds)	3.73 ± 0.49	3.11 ± 0.61
WavLM-base+ full fine-tune (3 seeds)	1.69 ± 0.06	1.41 ± 0.10

3. AS-Norm [7] with top- $K = 300$ cohort statistics on a 500-speaker in-domain cohort, layered on top of either backend (1) or (2).

3.6 Per-language and pooled evaluation

EER and MinDCF (with $P_{\text{target}} = 0.05$, $C_{\text{miss}} = 1$, $C_{\text{fa}} = 1$) are reported per trial list (Si, Ta, Cs) and on the concatenation. The trainer evaluates all four operating points in a single invocation. We report two EER definitions: the conservative $\max(\text{FPR}, \text{FNR})$ form historically used by the upstream toolkit and the standard literature $(\text{FPR} + \text{FNR})/2$ form. We use the latter as the primary metric to align with NIST SRE convention.

3.7 Reproducibility protocol

All experiments use the deterministic-mode switch (cuDNN deterministic, TF32 disabled, `torch.use_deterministic_algorithms` enabled). Primary configurations are run with three random seeds (42, 123, 7); we report mean $\pm$ std. Single-feature ablations use seed 42 only. Per-feature contribution is measured as ΔEER when the named feature is disabled from the full stack.

4 Experimental results

4.1 Preliminary monolingual baselines (benchmark v0)

Table 1 reports preliminary monolingual results on benchmark v0. Two findings shape the cross-lingual study: (i) full fine-tuning beat LoRA at every seed on this closed-set benchmark, contrary to the published PEFT-beats-full-FT prediction for language shift — the v1 open-set rerun arbitrates; and (ii) per-language calibration is mandatory: swapping a calibrator fitted on one language onto the other inflates Cllr by up to $5.7\times$. The cross-lingual (Cs) trial columns below remain pending: they require the bilingual speakers that only benchmark v1 (SLCeleb) provides.

4.2 Primary configurations

4.3 Single-feature ablations

4.4 Per-language characterisation

[INSERT: per-language EER comparison figure — bar chart with three groups (Si / Ta / Cs) and three bars per group (P0 / P1 / P2). The expected pattern: Cs EER is the highest of the three; the Si-Ta gap shrinks as features are added.]

Table 2: Equal-error rate (%) on the SL-SPV evaluation set, mean $\pm$ std over three random seeds.

Config	Pooled	Si	Ta	Cs
P0 from scratch	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]
P1 + fine-tune	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]
P2 full stack	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]	[X.X $\pm$ Y.Y]

Table 3: Single-feature ablations from the full stack (P2), seed 42. Δ = pooled-EER increase when the named feature is removed.

Ablation	Pooled EER %	Δ
P2 (all features)	[X.X]	0
– fine-tune	[X.X]	+ [Y.Y]
– LLRD	[X.X]	+ [Y.Y]
– lang-aux	[X.X]	+ [Y.Y]
– AS-Norm	[X.X]	+ [Y.Y]
– PLDA	[X.X]	+ [Y.Y]

4.5 Cross-lingual EER gap

[INSERT: discussion of the magnitude of (Cs – pooled) EER, which is the headline number for cross-lingual robustness. Smaller is better. Compare across P0 / P1 / P2 to characterise which interventions help close the cross-lingual gap.]

5 Discussion

5.1 Cross-lingual transfer is the dominant lever

[NOTE: discuss after results land. Expected pattern: the largest Δ in Table 3 is the fine-tune ablation, by a wide margin, indicating that cross-lingual transfer from English contributes more than any in-domain regularisation technique.]

5.2 AS-Norm and PLDA composition

[NOTE: discuss whether the AS-Norm contribution remains positive when PLDA is on, or whether the two redundantly model the same residual variance.]

5.3 Limitations

Benchmark v0, while larger than initially planned (527 speakers), is read speech from a single source per language and near single-session per speaker, so absolute EERs are optimistic and channel diversity is limited; its fine-tuning evaluation is also closed-set (utterance-disjoint but speaker-overlapping). Benchmark v1 (SLCeleb, 280 in-the-wild, multi-session speakers) addresses both. Continued self-supervised pretraining on unlabelled SL audio — shown in the literature to add 5–15% relative EER on top of off-the-shelf multilingual SSL — is documented in the toolkit as infrastructure-ready future work (FEATURE-009).

6 Conclusion

We presented a reproducible two-tier speaker-verification benchmark for Sri Lankan Sinhala and Tamil (v0: 527 read-speech speakers from OpenSLR SLR52/SLR65; v1: SLCeleb, our 280-speaker in-the-wild corpus, pending), the first such benchmark to our knowledge. Single-feature ablations isolate the contributions of cross-lingual fine-tuning, layer-wise LR decay, multi-task language-ID, AS-Norm, and PLDA. The open-source toolkit, corpus preparation pipeline, and reproducibility checklist are released to support downstream low-resource SV research, particularly on the languages of the South Asian region for which no published baselines currently exist.

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Declarations

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Competing interests. The authors have no competing interests to declare.

Data availability. The corpus-preparation pipeline and trial lists are publicly released alongside this paper; the raw corpus is available subject to [statement about data licensing / ethics-board approval].

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